



AZURE DATA ENGINEERING

About KSR Datavizon

Welcome to KSR Datavizon, we are your one-stop solution for all trending technologies. We provide in-depth knowledge about all trending courses with 100% placement guaranteed. You also will get hands-on demonstrations, interactive live sessions to learn and grow more on current trending courses like AI, Machine Learning, Data Science, Cloud Computing, Full stack Development, Cyber Security, Credit Risk modelling, DevOps, Azure, AWS, PowerBi, Web Development, Data Engineer, ADF, MSBI and Snowflakes and many more

You also get career guidance, soft skills sessions and resume building and all other interview tips to get settled in your dream career.

Getting Started with Data Engineering

About the Programme

Data engineering is the branch of data science that focuses on practical applications of data analysis and collection. Like other branches of engineering, data engineering deals with applying data science in the real world. Data engineering isn't related to experimental design. It is more focused on developing systems for better flow and access to the information.

This course covers the Hadoop, Spark, Kafka key concepts, architectures, features, considerations and best practices intended for key stakeholders who will be accessing, developing, querying datasets for analytic tasks and building data pipelines in Spark.



Course Curriculum

ADVANCED SQL / NoSQL



ADVANCED PYTHON



DATA WAREHOUSING



BIG DATA / HADOOP / SPARK



**INTRODUCTION TO CLOUD
COMPUTING**



Azure



AZURE SERVICES



Modules :



MySQL

Advanced MySQL



- SQL Advanced Queries
- SQL Data Models
- SQL Window Functions
- SQL Analytical Functions
- SQL CTE / Hierarchical Queries
- Introduction to NoSQL
- SQL vs. NoSQL
- Types of NoSQL



Python

Introduction



- History & features of Python
- Python Vs Other Programming Languages
- First Python Program
- Python basic syntax
- Python Development Tools & Packages

Python Environment Setup



- Installing Python
- Verify & Setup Python environment

Python Data Types



- Variables
- Data Types
- Strings
- Type Casting

Python Operators



- Arithmetic Operators
- Relational Operators
- Logical Operators
- Bitwise Operators
- Assignment Operator

Modules :



Python Flow Control & Loops



- if, if else, if else if
- while, do while loops
- for loops

Python Functional Programming



- Function Declarations
- Calling Functions
- Functions Call-by-Name
- Functions with Named Arguments
- Functions with Variable Arguments
- Functions with Default Parameter Values
- Lambda Functions

Python Collections



- Lists
- Tuples
- Sets
- Dictionaries

Python Files I/O



- Reading input from console
- Reading data from Files
- Writing data to Files

Python Object Oriented Programming



Python Classes

- Simple class
- Class objects
- Inheritance

Python Exception Handling



- Throwing Exceptions
- try, catch, finally
- Catching Exceptions
- The Finally Clause Python Miscellaneous
- Modules
- Dates
- RegEx



Data Warehouse

Introduction



- What is Data Warehouse?
- Why do you need Data Warehouse?
- Ideal Data Warehouse Solution

Architecture

OLAP Vs. OLTP

Database Schema's



- Star Schema
- Snowflake Schema

Data Modelling



- Fact Tables
- Dimension Table
- Fact less Table

ETL



- Introduction to ETL
- Compare ETL to ELT
- ETL Design





Big Data / Hadoop / Spark

Bigdata



- Introduction to Bigdata
- Growth of Bigdata

Bigdata Problems



- Traditional Database problems
- Traditional Database Processing problems

Hadoop as Solution

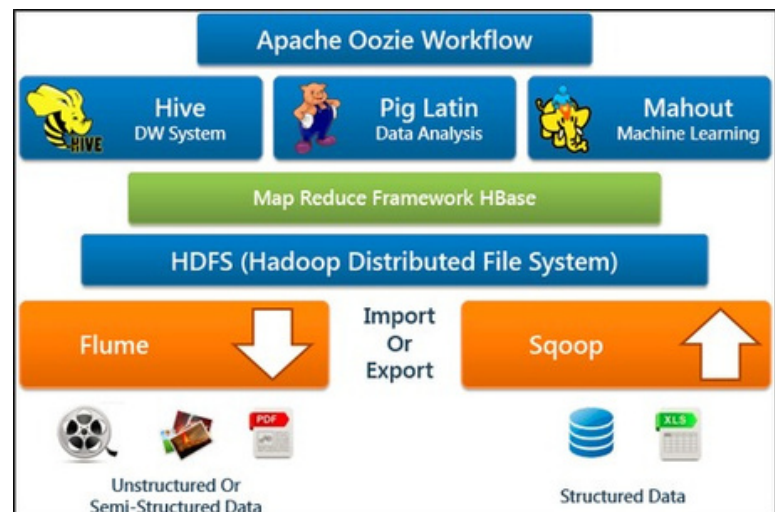


- Hadoop Architecture
- HDFS and MapReduce
- Hadoop 1.X and Hadoop 2.X
- HIVE, PIG, MapReduce

Limitations of Hadoop

Spark

- Introduction of Spark
- Spark Architecture
- PySpark





Cloud Computing

Introduction to Cloud Computing



- Problems in Traditional Storage Systems
- Problems in Traditional Processing Services
- Cloud Computing as Solution
- a. IAAS
- b. PAAS
- c. SAAS
- Cloud Services
- Comparison of AZURE, AWS and GCP

AZURE



- Overview of AZURE
- AZURE Free Account
- Azure Portal
- Azure Services and Applications



AZURE

Azure Storage



- Azure Blob Storage
- Azure Data Lake Store Gen I & Gen II
- Azure Disk Storage

Azure SQL Database



- Introduction to Azure SQL Database
- Azure Single Database

Azure Data Factory



- What is Data Factory?
- Data Factory Components
- Data Factory – Pipelines
- Data Factory – Activities
- Data Factory – Linked Services
- Data Factory – Datasets
- Data Factory – Triggers
- Data Factory – Dataflow
- Data Factory – Parameters
- Data Factory – Monitoring

AZURE SERVICES

Azure Synapse Analytics



- What is Synapse Analytics Service?
- b. Create Dedicated SQL Pool Explore Synapse Studio V2
- c. Analyse Data using Apache Spark Notebook
- d. Analyse Data using Dedicated SQL Pool
- e. Monitor Synapse Studio

Azure Event Hub, IoT Hub

Azure Stream Analytics



- Introduction to Azure Streaming Analytics Service
- Spark Analytics - Time Windowing
- Spark Analytics - Monitoring
- Spark Analytics - Optimization
- Spark Analytics - Types of Window Functions



Azure Data Bricks

Azure Databricks



- Introduction of Databricks
- Why Databricks in Cloud?
- Azure - Databricks Clusters
- Azure - Databricks Components
- Azure - Monitoring
- Explore, Analyse, Clean, Transform and Load Data in Databricks

Azure Cosmos DB



- Introduction to Cosmos DB
- Cosmos DB Features
- Storage vs. Cosmos DB
- Provision Cosmos DB Account
- Cosmos - Database, Containers and items

PROJECT



- On-Premise ETL Demo Example
- Cloud ETL Demo Example
- On-Premise Databases Migration
- End to End Project - Health Care



WHAT YOU WILL GET AFTER COMPLETION OF THIS COURSE



**Course Completion
certificate**



**Building an impressive
LINKEDIN and
NAUKRI Profile**



**Client handling
Techniques**



**Resume Forwarding to
Recruitment Team & Support
till you get Interview Calls**



**Building an Impressive
ATS Friendly Resume**



**Interview
Preparation Tips and
Mock Interviews**



**Interview Preparation Tips and
Mock Soft Skills and Corporate
Office Etiquette's**



**LMS Platform Access for
365 Days (All Recorded
Sessions)**



**Attend 1 Week Free
Sessions**



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Classes**



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